

Assignment 5 — Math Fundamentals for Robotics 16-811, Fall 2024

Mukai Yu
MSR, CMU
Pittsburgh, PA
mukaiy@andrew.cmu.edu

1. Consider a plane curve $y(x)$ over the interval $[x_0, x_1]$, with specified endpoints $y_0 = y(x_0)$ and $y_1 = y(x_1)$. Assume that $y_0 > 0$ and $y_1 > 0$ and that $y(x) \geq 0$ for $x_0 \leq x \leq x_1$. Now imagine rotating the curve about the x-axis to obtain a surface of revolution. Find the C^2 curve $y(x)$ with specified endpoints that minimizes the surface area of this surface of revolution.

[Hint: This problem explores further some of the limitations of the Calculus of Variations. Depending on the endpoint conditions there may or may not be a C^2 solution. What does the optimal “curve” look like when it is not C^2 ? Can you say how the endpoint conditions matter? Be aware: There are many subtleties; don’t expect to cover all, but explore what you can.]

Solution:

Surface area of revolution is given by

$$S = \int_{x_0}^{x_1} 2\pi y(x) \sqrt{1 + (y'(x))^2} dx.$$

We want to minimize S subject to the constraints $y(x_0) = y_0, y(x_1) = y_1$, so the optimization program is

$$\begin{aligned} \min_y J(y) &= \int_{x_0}^{x_1} F(x, y, y') dx, \\ \text{s.t. } &\begin{cases} y_0 &= y(x_0) \\ y_1 &= y(x_1) \end{cases} \end{aligned}$$

Where $F(x, y, y') = 2\pi y(x) \sqrt{1 + (y'(x))^2}$.

Now suppose y^* minimizes J , and $\eta \in C^2[x_0, x_1]$ be a random Δ function s.t. $\eta(x_0) = \eta(x_1) = 0$. And we define $y(x) = y^*(x) + \epsilon \eta(x)$, where $\epsilon \in \mathbb{R}$ is a small number. Then we can define $J(S)$ w.r.t. ϵ as

$$\begin{aligned} \Phi(\epsilon) &= \int_{x_0}^{x_1} F(x, y, y') dx \\ &= \int_{x_0}^{x_1} F(x, y^*(x) + \epsilon \eta(x), y^{*'}(x) + \epsilon \eta'(x)) dx. \end{aligned}$$

And we can calculate $\Phi'(0)$ as

$$\begin{aligned} \Phi'(0) &= \int_{x_0}^{x_1} \frac{\partial F}{\partial y} \eta(x) dx + \int_{x_0}^{x_1} \frac{\partial F}{\partial y'} \eta'(x) dx \\ &= \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx. \end{aligned}$$

Because y^* minimizes J , we have $\Phi'(0) = 0$. Also because η is arbitrary, so we must have the Euler-Lagrange equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

Because we have

$$\begin{cases} \frac{\partial F}{\partial y} &= 2\pi \sqrt{1 + (y'(x))^2} \\ \frac{\partial F}{\partial y'} &= 2\pi y(x) \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} \end{cases}$$

We can simplify the Euler-Lagrange equation as

$$\begin{aligned}
\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) &= 2\pi \sqrt{1 + (y'(x))^2} - \frac{d}{dx} \left(2\pi y(x) \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} \right) \\
&= 2\pi \sqrt{1 + (y'(x))^2} - 2\pi \frac{d}{dx} \left(y(x) \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} \right) = 0 \Rightarrow \\
\frac{d}{dx} \left(\frac{y(x)y'(x)}{\sqrt{1 + (y'(x))^2}} \right) &= \sqrt{1 + (y'(x))^2}
\end{aligned}$$

Because $F(x, y(x), y'(x))$ does not explicitly depend on x , we can use the **Beltrami Identity** to simplify the equation as

$$\begin{aligned}
F - y'(x) \frac{\partial F}{\partial y'} &= C = \text{constant} \Rightarrow \\
2\pi y(x) \sqrt{1 + (y'(x))^2} - 2\pi y'(x) \frac{y(x)y'(x)}{\sqrt{1 + (y'(x))^2}} &= 2\pi \frac{y(x)}{\sqrt{1 + (y'(x))^2}} = C \Rightarrow \\
y'(x) &= \pm \sqrt{(Cy)^2 - 1}
\end{aligned}$$

Note that the constant C is absorbing all the constants in the equation, so we can solve the equation by integrating both sides as

$$\int \frac{dy}{\sqrt{(Cy)^2 - 1}} = \pm \int dx.$$

Then we get

$$y(x) = \frac{\cosh(C(x - x_0))}{C}$$

Where C depends on the boundary conditions (x_0, y_0) and (x_1, y_1) .

The shape visualized with Desmos looks like a thin vase with big top and bottom, which is reasonable since we want the shape to shrink a little before it tries to reach another boundary, in order to minimize the surface area.

2. Using Calculus of Variations, show that the shortest curve between two points on a sphere is an arc of a great circle. [Hints: There are different ways to solve this problem. Here is one approach: Use spherical (u, v) coordinates, where $x = R \sin v \cos u, y = R \sin v \sin u, z = R \cos v$, with R the radius of the sphere. (Do not worry about singularities in the representation.) Rephrase 3D arclength $\sqrt{dx^2 + dy^2 + dz^2}$ as a function of u, v, du, dv . View the curve as a function $v(u)$. Observe that u does not appear directly in the resulting integrand for arclength, so replace the Euler-Lagrange equation with another, as in lecture. You may find the following identity useful:

$$\int \frac{a}{\sqrt{\sin^4 w - a^2 \sin^2 w}} dw = -\sin^{-1}\left(\frac{\cot w}{\sqrt{\frac{1}{a^2} - 1}}\right) + k,$$

where a and k are appropriate constants.]

Solution:

The arclength of a curve on a sphere is given by

$$L = \int_{u_0}^{u_1} R \sqrt{[v'(u)]^2 + \sin^2 v(u)} du.$$

Now remove constant linear parameter R and let $F(u, v(u), v'(u)) = \sqrt{[v'(u)]^2 + \sin^2 v(u)}$. Omitting all the derivation we did in Question 1, the Euler-Lagrange equation is

$$\frac{\partial F}{\partial v} - \frac{d}{du} \left(\frac{\partial F}{\partial v'} \right) = 0.$$

And we have

$$\begin{cases} \frac{\partial F}{\partial v} = \frac{\sin v(u) \cos v(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} \\ \frac{\partial F}{\partial v'} = \frac{v'(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} \end{cases}$$

Similarly with **Beltrami Identity**, we can simplify the Euler-Lagrange equation as

$$\begin{aligned} F - v'(u) \frac{\partial F}{\partial v'} &= C = \text{constant} \Rightarrow \\ \sqrt{[v'(u)]^2 + \sin^2 v(u)} - v'(u) \frac{v'(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} &= C \Rightarrow \\ \sin^2 v(u) &= C \sqrt{[v'(u)]^2 + \sin^2 v(u)} \Rightarrow \\ [v'(u)]^2 &= \frac{\sin^4 v(u)}{C^2} - \sin^2 v(u) \Rightarrow \\ v'(u) &= \pm \sqrt{\frac{\sin^4 v(u)}{C^2} - \sin^2 v(u)} \end{aligned}$$

Separate variables and integrate both sides, we get

$$\int \frac{C^2 dv(u)}{\sqrt{\sin^4 v(u) - C^2 \sin^2 v(u)}} = \pm \int du.$$

Using the identity from hint, we have

$$\begin{aligned} -\sin^{-1} \left(\frac{\cot v(u)}{\sqrt{\frac{1}{C^2} - 1}} \right) &= \pm u + C' \Rightarrow \\ \cot v(u) &= -\sin(\pm u + C') \sqrt{\frac{1}{C^2} - 1} \Rightarrow \\ \tan v(u) &= -\frac{\frac{C}{\sqrt{1-C^2}}}{\sin(\pm u + C')} \end{aligned}$$

Absorbing the $\pm$ sign and rearrange constants, we can get

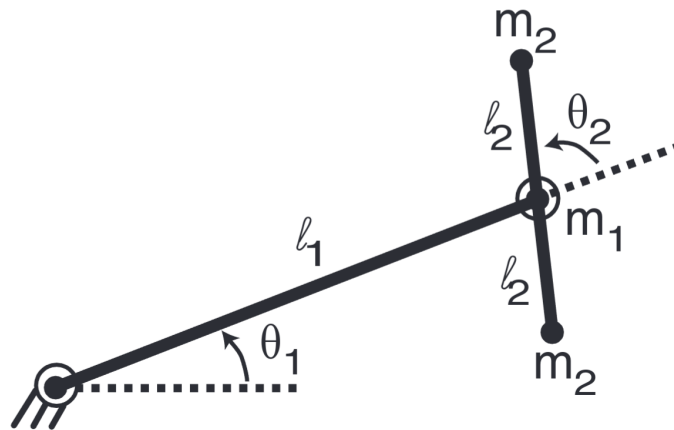
$$v(u) = \arctan \left(\frac{C_1}{\sin(u + C_2)} \right)$$

Where C_1, C_2 are constants depending on the boundary conditions (u_0, v_0) and (u_1, v_1) .

The shape visualized with Desmos looks like a periodic function with a period of 2π , where C_1 controls amplitude C_2 controls phase. This is reasonable since u is the longitude angle ranging $[0, 2\pi]$, and v is the latitude angle ranging $[0, \pi]$, so the curve should be periodic with a period of 2π since it's a function of u .

3. In the brachistochrone problem (“ski-slope” in lecture), suppose the right endpoint (x_1, y_1) is not specified exactly, but is merely constrained to satisfy an equation of the form $g(x, y) = 0$. Show that the optimizing curve $y(x)$ is perpendicular to the iso-contour $g(x, y) = 0$ at (x_1, y_1) .
[Hints: (i) Use an equation from lecture rather than derive everything from scratch. (ii) Observe that the curves are written differently, one explicitly as $y(x)$, the other implicitly as $g(x, y) = 0$. Think about how to compare the slopes of the two curves despite these different representations.]

- 4.(a) Using Lagrangian Dynamics, derive the relationship between joint torques and the angular state (angles, angular velocities, and angular accelerations) of the following balanced manipulator:



There is no gravity (in practice, gravity is perpendicular to the sheet of the paper).

Legend: All of link #1's mass, m_1 , is concentrated at distance l_1 from its rotational joint (which is attached to the ground). In turn, link #2 rotates around this distal point, with two masses, m_2 , located symmetrically, each at distance l_2 , from the joint. In practice, these two masses might constitute one counter-balanced end-effector or two different but equally weighted end-effectors. — This is a variation of a basic SCARA-type robot arm, used in industrial assembly.

- (b) When $\ddot{\theta}_2 = 0$, explain the terms relating $\ddot{\theta}_1$ to τ_1 .